SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1715**

COURSE OUTCOME COVERED

CO3: *Analyse knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 40%

Duration :1 Hour

QUESTIONS: 4 **Total Marks:40**

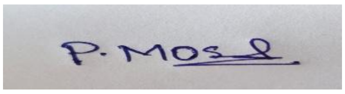
Annexure A

CASE STUDY

Code of Conduct:

I P.Moshiya(192572142) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

I. If a patient has Fever AND Cough → Possible Flu

Ii. If a patient has Fever AND Rash → Possible Measles

Iii. If a patient has Cough AND Breathlessness → Possible Pneumonia

Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

I. ∀x: Vehicle(x) ∧ Emergency Type(x) → ClearPath(x) [Rule 1]

II. ∀x: ClearPath(x) → Green Signal(x) ∧ ¬Red Signal(x) [Rule 2]

III. ∀x ∀y: Vehicle(x) ∧ Behind(X,Y) ∧ Green Signal(y) → Proceed(x) [Rule 3]

IV. Facts: Vehicle(Ambulance), Emergency Type(Ambulance), Vehicle(Car A), Behind(Car A, Ambulance)

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyse whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

1. P1: Soil Dry → Irrigation Needed
2. P2: Irrigation Needed ∧ Crop Wheat → ApplyDripMethod
3. P3: ¬ApplyDripMethod ∧ Crop Wheat → CropAtRisk
4. Facts: Soil Dry = True, Crop Wheat = True

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication → move ¬ inward → distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause □. Draw the resolution proof tree.

(c) If the fact 'Soil Dry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

1. Stench [1,2] → Wumpus is adjacent to cell [1,2]
2. Breeze [1,1] → Pit is adjacent to cell [1,1]
3. Glitter [x,y] → Gold is at [x,y]
4. ¬Wumpus [x,y] ∧ ¬Pit[x,y] → Safe[x,y]

(a) Construct the agent's belief state from its precepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's precepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] → [1,2] → [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyse the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Criteria	Weightage	Q1 (10)	Q2 (10)	Q3 (10)	Q4 (10)	Total Marks (40)
Understanding of Case	25% (2.5)	/2.5	/2.5	/2.5	/2.5	/10
Application of Concepts	25% (2.5)	/2.5	/2.5	/2.5	/2.5	/10
Analysis	20% (2)	/2	/2	/2	/2	/8
Solution Design	20% (2)	/2	/2	/2	/2	/8
Clarity	10% (1)	/1	/1	/1	/1	/4
Total per Question	**10 Marks**	**/10**	**/10**	**/10**	**/10**	**/40**

100

n

SOLUTION:

Case Study 1 – Smart Medical Diagnosis System

(a) Represent the rules and facts using Propositional Logic

Step 1: Define Symbols

Symbol	Meaning
F	Patient has Fever
C	Patient has Cough
R	Patient has Rash
B	Patient has Breathlessness
Flu	Patient has Possible Flu
M	Patient has Possible Measles
P	Patient has Possible Pneumonia

Step 2: Knowledge Base (KB)

Rule I:

If a patient has Fever AND Cough, then Possible Flu.

Rule II:

If a patient has Fever AND Rash, then Possible Measles.

Rule III:

If a patient has Cough AND Breathlessness, then Possible Pneumonia.

Facts about Patient A

- Fever = True
- Cough = True
- Breathlessness = True

(b) Apply Forward Chaining

Initial Facts

Known facts:

- Fever (F)
- Cough (C)
- Breathlessness (B)

Working Memory:

Step 1

Check Rule I

Both F and C are TRUE.

Therefore,

Infer

Working Memory becomes

Step 2

Check Rule II

Need Rash (R).

But Rash is NOT given.

Therefore,

Rule II **cannot fire.**

Step 3

Check Rule III

Both C and B are TRUE.

Therefore,

Infer

Working Memory becomes

No More Rules Can Fire

Forward Chaining Stops.

Inference Table

Step	Rule Fired	New Fact Derived
Initial Facts		F, C, B

Step	Rule Fired	New Fact Derived
1	F ∧ C → Flu	Flu
2	F ∧ R → M	Cannot Fire
3	C ∧ B → P	Pneumonia

Final Diagnoses

Possible Flu

Possible Pneumonia

Possible Measles (cannot be concluded because Rash is absent)

(c) Apply Backward Chaining

Goal

Prove

(Patient A has Pneumonia)

Step 1

Goal:

Look for a rule whose conclusion is P.

Found:

Now reduce the goal into two sub-goals.

Need to prove:

- C
- B

Step 2

Check Fact

Is C true?

Yes.

Success.

Step 3

Check Fact

Is B true?

Yes.

Success.

Since both sub-goals are satisfied,

Infer

Goal Achieved.

Goal Reduction Tree

```
     Goal: Pneumonia (P)
              |
   ----------------------
   |                |
 Cough (C)       Breathlessness (B)
   |                |
 Fact True         Fact True
   \________________________/
              |
      Pneumonia Proven
```

Backward Chaining Steps

Goal	**Rule Used**	**Result**
P	C ∧ B → P	Need C and B
C	Fact	True
B	Fact	True
P	Derived	Goal Achieved

Case Study 2 – Autonomous Traffic Management Agent

Given Knowledge Base

Rule 1

Rule 2

Rule 3

Facts

- Vehicle(Ambulance)
- Emergency Type(Ambulance)
- Vehicle(Car A)
- Behind(Car A, Ambulance)

(a) Apply Unification

Step 1: Rule 1

Rule:

Facts:

- Vehicle(Ambulance)
- Emergency Type(Ambulance)

Step 2: Match Predicate

Compare

Vehicle(x)

with

Vehicle(Ambulance)

Substitution

Step 3: Match Second Predicate

Emergency Type(x)

with

Emergency Type(Ambulance)

Apply previous substitution.

Again,

Most General Unifier (MGU)

Step 4: Apply Substitution

Rule becomes

Vehicle(Ambulance)

AND

Emergency Type(Ambulance)

→

ClearPath(Ambulance)

Both premises are true.

Hence derive

Answer (a)

MGU (θ):

Derived Fact:

(b) Resolution Method

Goal

Prove

Step 1: Convert Rules into CNF

Rule 1

Vehicle(x) ∧ Emergency Type(x) → ClearPath(x)

Remove implication

Rule 2 (Split into two clauses)

ClearPath(x) → Green Signal(x)

ClearPath(x) → ¬Red Signal(x)

Rule 3

Vehicle(x) ∧ Behind(x,y) ∧ Green Signal(y)

→ Proceed(x)

CNF

Facts

Vehicle(Ambulance)

Emergency Type(Ambulance)

Vehicle(Car A)

Behind(Car A, Ambulance)

Negated Goal

Goal:

Proceed(Car A)

Negate it.

Resolution Steps

Step 1

Resolve Rule 1

with

Vehicle(Ambulance)

and

Emergency Type(Ambulance)

Obtain

Step 2

Resolve

ClearPath(Ambulance)

with Rule 2

Obtain

Step 3

Use Rule 3

Vehicle(Car A)

Behind(Cara, Ambulance)

Green Signal(Ambulance)

All are true.

Therefore,

Step 4

Resolve

Proceed(Car A)

with

¬Proceed(Car A)

Result

(Empty Clause)

Hence theorem is proved.

Resolution Proof Tree

Vehicle(Ambulance)

Emergency Type(Ambulance)

|

|

Rule 1

|

ClearPath(Ambulance)

|

Rule 2

|

Green Signal(Ambulance)

|

Vehicle(Car A)

Behind(Car A, Ambulance)

|

Rule 3

|

Proceed(Car A)

|

¬Proceed(Car A)

|

□

Answer (b)

CNF Clauses

1.

2.

3.

4.

5.

Vehicle(Ambulance)

6.

Emergency Type(Ambulance)

7.

Vehicle(Car A)

8.

Behind(Cara, Ambulance)

9.

¬Proceed(Car A)

Resolution derives the empty clause □, proving that:

(c) Analysis

Is the current KB sufficient for handling two emergency vehicles?

No.

The current knowledge base is designed to reason about **only one emergency vehicle at a time**. If two emergency vehicles (for example, an Ambulance and a Fire Truck) approach the same intersection simultaneously, the KB cannot decide which one should get priority.

Additional Facts Needed

- Vehicle(Fire Truck)
- Emergency Type(Fire Truck)

Additional Rules Needed

1. **Priority Rule**

OR

2. **Conflict Resolution Rule**

If two emergency vehicles arrive together, the vehicle with the higher priority gets the green signal first.

3. **Sequential Signal Rule**

After the first emergency vehicle crosses,

where is the second emergency vehicle.

Justification

These additional rules help the system:

- Avoid traffic conflicts.
- Decide which emergency vehicle proceeds first.
- Ensure all emergency vehicles are routed safely and efficiently.

Thus, the knowledge base becomes capable of handling **multiple simultaneous emergency vehicles** logically.

Case Study 3 – Agricultural Expert Reasoning System

Given Knowledge Base (KB)

P1:

soil Dry → Irrigation Needed

P2:

Irrigation Needed ∧ Crop Wheat → Apply Drip Method

P3:

¬Apply Drip Method ∧ Crop Wheat → CropAtRisk

Facts:

- Soil Dry = True
- Crop Wheat = True

(a) Convert all sentences into Conjunctive Normal Form (CNF)

Step 1: Define Symbols

Symbol	**Meaning**
S	Soil Dry
I	Irrigation Needed
W	Crop Wheat
D	Apply Drip Method
R	CropAtRisk

P1: soil Dry → Irrigation Needed

Original

Remove implication

CNF

P2: Irrigation Needed ∧ Crop Wheat → Apply Drip Method

Original

Remove implication

Move negation inward (De Morgan's Law)

Rearrange

P3: ¬Apply Drip Method ∧ Crop Wheat → CropAtRisk

Original

Remove implication

Move negation inward

CNF

Facts

soil Dry=True

Crop Wheat=True

Final CNF Clauses

1.
2.
3.

4.

5.

(b) Resolution Refutation

Goal

Prove

Step 1: Negate Goal

Goal:

Negated Goal

Step 2: Add Negated Goal to KB

Clauses

1.

2.

3.

4.

5.

6.

Resolution Steps

Step 1

Resolve

with

Result

Step 2

Resolve

with

Result

Step 3

Resolve

with

Result

Step 4

Resolve

with

Result

(Empty Clause)

Hence,

Goal proved.

Resolution Proof Tree

S

|

¬S ∨ I

|

I

|

¬I ∨ ¬W ∨ D

|

¬W ∨ D

|

W

|

D

|

¬D

|

□

Answer (b)

Since the empty clause (□) is obtained,

(c) Analysis

Suppose Soil Dry is removed.

Remaining Facts

- Crop Wheat=True

Only

Step 1

Rule P1

cannot fire because

S is missing.

Therefore,

Irrigation Needed cannot be derived.

Step 2

Rule P2

Needs

Irrigation Needed

AND

Crop Wheat

Since Irrigation Needed is not available,

Apply Drip Method cannot be derived.

Step 3

Rule P3

Needs

¬Apply Drip Method

AND

Crop Wheat

We only know that Apply Drip Method cannot be proved.

This **does not mean** ¬Apply Drip Method is true.

In classical logic, failure to prove D is **not** the same as proving ¬D.

Therefore,

CropAtRisk **cannot** be concluded.

Reasoning Summary

Situation	**Result**
Soil Dry present	Irrigation Needed derived
Irrigation Needed + Crop Wheat	Apply Drip Method derived

Situation	Result
Apply Drip Method	Goal proved
Soil Dry removed	Irrigation Needed not derived
Apply Drip Method not derived	CropAtRisk cannot be concluded

Case Study 4 – Wumpus World Knowledge Agent

Given Knowledge Base (KB)

Precepts

- Stench at **[1,2]**
- Breeze at **[1,1]**
- Glitter at **[2,2]**

Rules

Rule 1

Stench[1,2] → Wumpus is adjacent to cell [1,2]

Rule 2

Breeze[1,1] → Pit is adjacent to cell [1,1]

Rule 3

Glitter[x,y] → Gold[x, y]

Rule 4

¬Wumpus[x,y] ∧ ¬Pit[x,y] → Safe[x,y]

(a) Construct the Agent's Belief State and Apply Modus Ponens

Step 1: Define Symbols

Symbol	Meaning
S(1,2)	Stench at cell [1,2]
B(1,1)	Breeze at cell [1,1]
G(2,2)	Glitter at cell [2,2]
W(x,y)	Wumpus at cell (x,y)
P(x, y)	Pit at cell (x ,y)
Gold(x ,y)	Gold at cell (x ,y)
Safe(x ,y)	Safe cell

Step 2: Initial Belief State

Known facts:

- S(1,2)
- B(1,1)
- G(2,2)

Step 3: Apply Modus Ponens

Rule 1

S(1,2) → Wumpus adjacent to (1,2)

Since **S(1,2)** is true,

Derived Fact:

Wumpus is adjacent to **[1,2]**

Possible adjacent cells:

- [1,1]
- [1,3]
- [2,2]

Rule 2

B(1,1) → Pit adjacent to (1,1)

Since **B(1,1)** is true,

Derived Fact:

Pit is adjacent to **[1,1]**

Possible adjacent cells:

- [1,2]
- [2,1]

Rule 3

G(2,2) → Gold(2,2)

Since **G(2,2)** is true,

Derived Fact:

Gold is located at **[2,2]**

Rule 4

¬Wumpus(x,y) ∧ ¬Pit(x,y) → Safe(x,y)

There is **no evidence** that any cell has **both no Wumpus and no Pit**, so no Safe cell can be concluded.

Derived Conclusions

- Wumpus is adjacent to **[1,2]**
- Pit is adjacent to **[1,1]**
- Gold is at **[2,2]**
- Safe cell **cannot be determined**

(b) Apply Forward Chaining

Initial Facts

- Stench(1,2)
- Breeze(1,1)
- Glitter(2,2)

Step 1

Apply Rule 1

Stench (1,2)

↓

Wumpus adjacent to (1,2)

Possible locations:

- [1,1]
- [1,3]
- [2,2]

Step 2

Apply Rule 2

Breeze(1,1)

↓

Pit adjacent to (1,1)

Possible locations:

- [1,2]
- [2,1]

Step 3

Apply Rule 3

Glitter(2,2)

↓

Gold(2,2)

Forward Chaining Result

Percept	**Rule Applied**	**Conclusion**
Stench(1,2)	Rule 1	Wumpus near [1,2]
Breeze(1,1)	Rule 2	Pit near [1,1]
Glitter(2,2)	Rule 3	Gold at [2,2]

Possible Locations

Wumpus

- [1,1]
- [1,3]
- [2,2]

Pit

- [1,2]
- [2,1]

Gold

- [2,2]

(c) Evaluate the Path

Path

Cell [1,1]

This is the starting cell.

A breeze is perceived here, which means a pit is nearby, **not necessarily in [1,1]**.

So the starting cell is considered safe.

Cell [1,2]

A stench is perceived here.

This indicates that the Wumpus is in one of the adjacent cells.

Also, based on the breeze at [1,1], **[1,2] could contain a pit**.

Therefore, **[1,2] is not confirmed to be safe**.

Cell [2,2]

Glitter indicates that **gold is present**.

However, from Rule 1, **[2,2] is also a possible location of the Wumpus**.

Thus, moving to **[2,2]** may be dangerous.

Conclusion

The path

cannot be considered safe because:

- **[1,2] may contain a pit.**
- **[2,2] may contain the Wumpus**, even though it contains the gold.

The agent requires additional precepts or reasoning before safely moving along this path.

Final Answers

(a) Belief State

Initial Facts:

- Stench(1,2)
- Breeze(1,1)
- Glitter(2,2)

Derived using Modus Ponens:

- Wumpus is adjacent to **[1,2]**
- Pit is adjacent to **[1,1]**
- Gold is at **[2,2]**
- No safe cell can be confirmed.

(b) Forward Chaining

- Stench(1,2) → Wumpus near **[1,2]**

- Breeze(1,1) → Pit near **[1,1]**
- Glitter(2,2) → Gold at **[2,2]**

Possible Wumpus locations: [1,1], [1,3], [2,2]

Possible Pit locations: [1,2], [2,1]

Gold location: [2,2]

(c) Path Safety

Path: **[1,1] → [1,2] → [2,2]**

- **[1,1]**: Starting cell (safe).
- **[1,2]**: May contain a pit.
- **[2,2]**: Contains gold but may also contain the Wumpus.